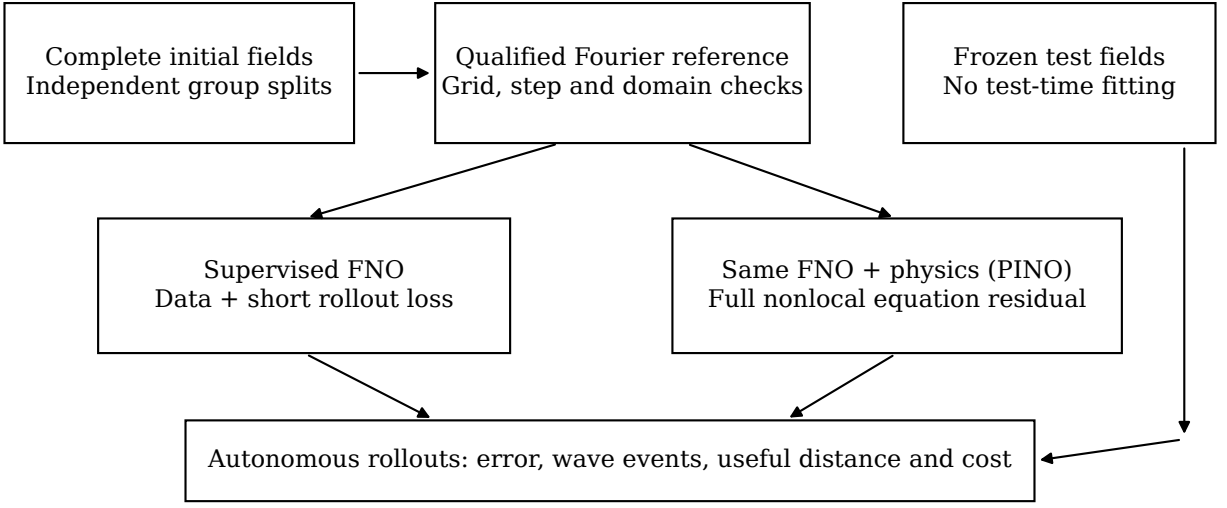
Complete initial fields
Independent group splits
Qualified Fourier reference
Grid, step and domain checks
Frozen test fields
No test-time fitting
Supervised FNO
Data + short rollout loss
Same FNO + physics (PINO)
Full nonlocal equation residual
Autonomous rollouts: error, wave events, useful distance and cost